

ELKINS RANDOLPH COUNTY AP, WV, USA

WMO: 724170

Lat: 38.890N

Lon: 79.855W

Elev: 1961

StdP: 13.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13729

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.2	7.4	-6.4	4.2	5.5	-1.0	5.6	10.7	21.8	37.5	19.4	36.3	2.3	270	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	86.4	70.5	84.4	69.5	82.5	68.5	73.2	82.6	72.0	80.7	70.8	79.1	6.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	119.8	77.6	69.1	115.0	76.3	68.0	110.5	75.3	38.0	82.5	36.8	80.9	35.7	79.0	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	15.9	13.0	DB	-9.0	90.3	5.9	2.8	-13.2	92.3	-16.6	93.9	-19.9	95.5	-24.2	97.5
			WB	-8.7	75.7	5.2	1.7	-12.4	76.9	-15.5	77.9	-18.4	78.8	-22.1	80.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	29.5	32.7	40.2	51.0	59.4	67.0	70.3	69.6	63.4	52.2	41.5	34.2
	DBStd	16.82	12.74	11.45	10.60	9.07	7.65	5.36	4.51	4.37	6.42	8.47	9.16	10.32
	HDD50	2443	644	492	336	97	15	0	0	0	2	77	283	497
	HDD65	5633	1100	904	769	423	203	43	11	11	104	405	706	954
	CDD50	2810	9	8	32	128	307	509	629	606	403	144	27	8
	CDD65	526	0	0	0	4	30	102	175	153	55	7	0	0
	CDH74	4591	0	1	14	135	401	880	1352	1176	532	96	4	0
	CDH80	1034	0	0	1	18	62	178	356	291	116	12	0	0
Wind	WSAvg	4.4	5.6	5.8	5.9	5.9	4.3	3.5	3.0	2.6	2.8	3.7	4.4	5.0
Precipitation	PrecAvg	44.4	3.1	3.0	4.0	3.8	4.3	4.3	4.8	4.1	3.9	2.9	3.2	3.4
	PrecMax	58.7	6.3	5.6	8.9	6.9	7.7	10.0	8.8	10.4	8.9	6.3	11.1	6.7
	PrecMin	34.1	1.1	0.8	1.5	1.0	1.5	1.7	1.3	1.1	0.3	0.3	0.9	0.9
	PrecStd	5.7	1.3	1.1	1.7	1.5	1.9	2.0	1.9	1.9	1.7	1.5	1.8	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.0	68.2	75.8	82.5	84.7	87.0	89.5	88.4	86.9	81.3	72.7	65.8
		MCWB	56.2	56.0	57.5	59.7	66.7	70.2	72.9	72.1	67.0	64.6	57.6	56.7
	2%	DB	60.4	61.2	69.8	78.0	81.7	84.3	86.4	85.8	83.4	76.4	67.1	60.7
		MCWB	53.3	50.7	54.6	58.9	64.7	69.5	71.8	71.1	67.3	62.0	54.6	53.4
	5%	DB	55.0	56.1	64.3	73.9	78.9	82.1	84.0	83.4	80.3	72.8	62.7	56.1
		MCWB	49.6	48.7	51.8	57.1	63.8	68.8	70.4	70.0	66.5	60.2	53.0	50.2
	10%	DB	49.1	51.2	58.8	69.4	75.4	79.6	81.6	80.9	76.7	68.5	58.5	51.5
		MCWB	44.1	44.6	49.6	54.7	62.4	67.5	69.4	69.0	64.9	58.4	50.7	46.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.3	57.6	59.3	63.8	69.6	73.3	75.2	75.0	72.1	68.4	60.8	58.2
		MCDB	63.3	64.7	72.2	76.2	80.4	82.7	85.4	84.5	81.5	77.6	68.8	64.1
	2%	WB	54.2	53.3	56.5	61.4	67.8	71.6	73.3	73.1	69.9	65.5	57.6	54.6
		MCDB	59.3	58.8	66.1	73.0	77.9	80.3	83.2	82.1	78.6	72.7	64.4	59.4
	5%	WB	50.3	49.7	54.1	59.4	66.0	70.2	72.1	71.7	68.4	62.5	54.8	51.3
		MCDB	54.6	55.4	61.6	69.6	75.0	79.4	81.0	80.1	76.3	69.4	60.3	55.5
	10%	WB	44.8	45.1	50.9	57.2	64.0	68.9	70.8	70.4	66.7	59.8	51.8	46.9
		MCDB	49.0	51.0	58.1	66.9	72.4	77.6	78.9	78.2	73.8	66.6	57.5	50.6
Mean Daily Temperature Range	5% DB	MDBR	19.2	20.7	22.7	26.0	23.7	21.8	21.0	21.3	22.8	24.5	22.8	19.2
		MCDBR	24.0	29.1	33.0	34.3	29.8	25.5	24.1	24.4	27.9	31.3	31.6	26.2
		MCWBR	17.5	20.1	18.9	16.8	14.6	12.6	10.9	11.2	13.5	16.8	20.4	19.2
	5% WB	MCDBR	21.1	25.2	26.9	26.0	23.9	21.8	20.8	20.6	22.1	24.2	24.2	22.6
		MCWBR	16.6	19.1	17.3	14.4	12.7	11.2	10.2	10.2	11.3	14.4	17.5	18.2
Clear-Sky Solar Irradiance	taub	0.292	0.312	0.336	0.377	0.417	0.445	0.464	0.477	0.382	0.346	0.310	0.290	
	taud	2.477	2.408	2.412	2.332	2.223	2.175	2.145	2.079	2.341	2.444	2.517	2.518	
	Ebn at Noon	282	290	293	285	274	266	259	252	273	273	273	275	
	Edh at Noon	25	30	34	39	44	47	48	50	36	29	24	23	
All-Sky Solar Radiation	RadAvg	570	822	1149	1500	1666	1845	1777	1636	1366	995	703	503	
	RadStd	62	119	82	124	144	125	107	86	159	119	79	50	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.03	N/A	N/A	N/A	N/A	N/A	N/A	-322	+315	+121
(47)	Regional (31 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+212	+104

Nomenclature: See separate page